# Supplement — When Compression Backfires (Full Revision R1)

TASCO Paper B: frozen-evidence detail for the main text

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## S-A: Calibers, pairing, and the noise floor

**Calibers.** *Input tokens*: non-cached, api-reported; primary. *Cache-read tokens*: cached re-send traffic. *Output tokens*. *Flow tokens* = input+output+reasoning+cache. *Wall time*, *tool calls*, *oracle success* (frozen test commands or deterministic text-token oracles), *fold/store events*, *recovery reads*. Character counts (saved/re-injected/marker/residual) are *opportunity proxies* and are never added to token quantities.

**Two input aggregates.** (a) *Totals-based* $\Delta = (\sum T - \sum B)/\sum B$ — weights large cases more; the M1 headline caliber (+7.1%). (b) *Case-equal-weight* mean of per-case $(B - T)/B$ with case-level bootstrap 95% CI (10,000 resamples, seed 20261019) — the M3 headline caliber (−23.0%, CI $[-40.8, -7.6]$). The two answer different questions; neither is "the" effect size, and we never mix them in one number.

**Hierarchy.** Independent unit = *task case*. M1/M2: 1 run per arm per case. M3: 2 runs per arm per case, aggregated to arm means *before* pairing (reps are repeated measures, not independent tasks). Stage-0: 8 tasks × 3 baseline reps, used only for the noise floor $\hat{\sigma}_w = 0.155$ (bootstrap 95% CI $[0.093, 0.197]$) and the median within-task paired swing 10.25%. Case-ID intersections across M1/M2/M3: 0 (verified programmatically; the four experiments are disjoint).

## S-B: Robustness detail

- **M1** (100 pairs): case-equal-weight mean −41.8% [CI $-66.7, -20.2$]; median −4.0%; win/loss/tie (input caliber) 42/58/0; leave-one-out mean range $[-42.9\%, -34.4\%]$; worst case C-ENT04 (benefit −774.7%; T input 8.7× B); best case C-ENT05-class enterprise case is included in the exploratory enterprise family.
- **M2** (30 pairs): mean −21.3% [CI $-50.3, +2.6$]; median +4.9%; win/loss/tie 17/13/0; leave-one-out mean range $[-23.9\%, -12.6\%]$; the mean is dominated by one exploratory outlier R-XS-AUTH-REP (−272.9%). Frozen flow-caliber median paired delta: −0.25%.
- **M3** (30 arm-mean pairs): mean −23.0% [CI $-40.8, -7.6$]; median −15.3%; win/loss/tie 8/22/0; leave-one-out: dropping the largest outlier (X-NO-F01) moves the mean by +17.2pp to −5.8%-equivalent (−17.16pp sensitivity as frozen); zero-exposure control subgroup (6 cases, no markers): mean +10.6%, median +2.4%.
- **Flow-caliber win/loss**: M1 35/62/3; M2 14/15/1; M3 5/25/0.
- **Oracle totals**: M1 81→80; M2 24→24; M3 18→20. Per-family counts in the main text Table 3.

## S-C: Full eligibility grid (64 configurations)

Replay semantics: the frozen folding primitive is always invoked (anchor registration, v2 semantics); gates only decide whether the folded text is adopted; diagnostic/target/ specification reads are always kept verbatim. Split: seed 20261017, family-stratified 70/30. All volumes are opportunity proxies.

| Config | Hold cov. | Hold chars | Exposure% | Blocked risk |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| R2/D0.0/M0 | 0.333 | 13,730 | 1.66 | 83 |
| R2/D0.0/M1000 | 0.267 | 12,028 | 1.45 | 83 |
| R2/D0.0/M2000 | 0.267 | 12,028 | 1.45 | 83 |
| R2/D0.0/M500 | 0.267 | 13,526 | 1.63 | 83 |
| R2/D0.2/M0 | 0.033 | 912 | 0.11 | 83 |
| R2/D0.2/M1000 | 0.033 | 912 | 0.11 | 83 |
| R2/D0.2/M2000 | 0.033 | 912 | 0.11 | 83 |
| R2/D0.2/M500 | 0.033 | 912 | 0.11 | 83 |
| R2/D0.3/M0 | 0.000 | 0 | 0.00 | 83 |
| R2/D0.3/M1000 | 0.000 | 0 | 0.00 | 83 |
| R2/D0.3/M2000 | 0.000 | 0 | 0.00 | 83 |
| R2/D0.3/M500 | 0.000 | 0 | 0.00 | 83 |
| R2/D0.4/M0 | 0.000 | 0 | 0.00 | 83 |
| R2/D0.4/M1000 | 0.000 | 0 | 0.00 | 83 |
| R2/D0.4/M2000 | 0.000 | 0 | 0.00 | 83 |
| R2/D0.4/M500 | 0.000 | 0 | 0.00 | 83 |
| R3/D0.0/M0 | 0.333 | 13,730 | 1.66 | 83 |
| R3/D0.0/M1000 | 0.267 | 12,028 | 1.45 | 83 |
| R3/D0.0/M2000 | 0.267 | 12,028 | 1.45 | 83 |
| R3/D0.0/M500 | 0.267 | 13,526 | 1.63 | 83 |
| R3/D0.2/M0 | 0.033 | 912 | 0.11 | 83 |
| R3/D0.2/M1000 | 0.033 | 912 | 0.11 | 83 |
| R3/D0.2/M2000 | 0.033 | 912 | 0.11 | 83 |
| R3/D0.2/M500 | 0.033 | 912 | 0.11 | 83 |
| R3/D0.3/M0 | 0.000 | 0 | 0.00 | 83 |
| R3/D0.3/M1000 | 0.000 | 0 | 0.00 | 83 |
| R3/D0.3/M2000 | 0.000 | 0 | 0.00 | 83 |
| R3/D0.3/M500 | 0.000 | 0 | 0.00 | 83 |
| R3/D0.4/M0 | 0.000 | 0 | 0.00 | 83 |
| R3/D0.4/M1000 | 0.000 | 0 | 0.00 | 83 |
| R3/D0.4/M2000 | 0.000 | 0 | 0.00 | 83 |
| R3/D0.4/M500 | 0.000 | 0 | 0.00 | 83 |
| R4/D0.0/M0 | 0.333 | 13,730 | 1.66 | 83 |
| R4/D0.0/M1000 | 0.267 | 12,028 | 1.45 | 83 |
| R4/D0.0/M2000 | 0.267 | 12,028 | 1.45 | 83 |
| R4/D0.0/M500 | 0.267 | 13,526 | 1.63 | 83 |
| R4/D0.2/M0 | 0.033 | 912 | 0.11 | 83 |
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| R4/D0.2/M2000 | 0.033 | 912 | 0.11 | 83 |
| R4/D0.2/M500 | 0.033 | 912 | 0.11 | 83 |
| R4/D0.3/M0 | 0.000 | 0 | 0.00 | 83 |
| R4/D0.3/M1000 | 0.000 | 0 | 0.00 | 83 |
| R4/D0.3/M2000 | 0.000 | 0 | 0.00 | 83 |
| R4/D0.3/M500 | 0.000 | 0 | 0.00 | 83 |
| R4/D0.4/M0 | 0.000 | 0 | 0.00 | 83 |
| R4/D0.4/M1000 | 0.000 | 0 | 0.00 | 83 |
| R4/D0.4/M2000 | 0.000 | 0 | 0.00 | 83 |
| R4/D0.4/M500 | 0.000 | 0 | 0.00 | 83 |
| R99/D0.0/M0 | 0.333 | 13,730 | 1.66 | 83 |
| R99/D0.0/M1000 | 0.267 | 12,028 | 1.45 | 83 |
| R99/D0.0/M2000 | 0.267 | 12,028 | 1.45 | 83 |
| R99/D0.0/M500 | 0.267 | 13,526 | 1.63 | 83 |
| R99/D0.2/M0 | 0.033 | 912 | 0.11 | 83 |
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| | | | | |
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| R99/D0.3/M0 | 0.000 | 0 | 0.00 | 83 |
| R99/D0.3/M1000 | 0.000 | 0 | 0.00 | 83 |
| R99/D0.3/M2000 | 0.000 | 0 | 0.00 | 83 |
| R99/D0.3/M500 | 0.000 | 0 | 0.00 | 83 |
| R99/D0.4/M0 | 0.000 | 0 | 0.00 | 83 |
| R99/D0.4/M1000 | 0.000 | 0 | 0.00 | 83 |
| R99/D0.4/M2000 | 0.000 | 0 | 0.00 | 83 |
| R99/D0.4/M500 | 0.000 | 0 | 0.00 | 83 |

Table 1: Full 64-configuration eligibility grid (holdout replay). Max exposure 1.66% (R2/D0.0/M0); no configuration reaches the $\geq$2% exposure $\wedge$ $\geq$1/3 coverage gates. Risk = blocked target/specification/diagnostic folds. All volumes are opportunity proxies.

## S-D: R6 recovery screening detail

Scope: all 70 treatment runs (60 expansion + 10 online smoke). Observed: 311 externalized views; 135 recovered at least once (43.4%); 137 recovery events; median first-recovery distance 4 tool events; re-injected 352,804 chars (per-run caliber) against 1,762,154 saved; 24 same-file rereads in 100 baseline runs without any side store. Per-view character nets (saved − recovered − marker): design +4,241, holdout +4,589. Feature separation: recovery rates by family (file-review 52.2%, module-analysis 37.6%), stratum (high 38.5%, normal, low 45.5%), size bucket, and extension all fall in the 38–52% band in both design and holdout splits — no pre-observable separating feature. Candidate screening verdict (frozen): no qualified candidate; the single stable positive case (X-HI-M01, zero recoveries, +23.9%) is identifiable only post hoc. The 24 baseline rereads show that "no recovery" $\neq$ "no need": demand can resurface as re-derivation rather than reads.

## S-E: G5 audit protocol and limited-sample detail

**Blind package.** 30 views (S01–S30) from 3 evidence runs (10/13/7; 3 underlying cases); each view = agent-visible in-context text + all attached side-store originals; frozen REVIEWER-SHEET (E1–E4 + PASS); sample hashes in the package inventory; the sealed mapping was never opened by the audit. **Dual-model protocol.** Two isolated reviewers (REVIEWER_A = DeepSeek-V4-Flash via the official Anthropic-compatible endpoint; REVIEWER_B = GLM-5.3-Flash via a Volcano-Engine-compatible gateway), frozen prompt template and schema, temperature 0, max_tokens 32768, stateless single-turn calls, symmetric parameters, retry only on format/transport failures (max 3 attempts), 92 API calls total, 60 valid units, response model fields verified per call. Verdicts: A 25 FAIL / 5 PASS; B 1 FAIL / 29 PASS; raw agreement 0.200, $\kappa = 0.0137$ (extremely imbalanced marginals; $\kappa$ not interpretable). **The verdicts were not used as evidence**: every reported defect was re-verified mechanically (below), and the reviewer disagreement was attributed to three pre-defined standard ambiguities (diagnostic-line definition; visibility vs. recoverability; marker coordinate system). **Mechanical verification (deterministic replay).** V1: stored original == repository file, line-for-line (30/30). V2: Python re-implementation of the frozen folding function reproduces every view byte-for-byte (30/30). V4: session event logs contain exactly the folded views (30/30 byte-identical). Findings: E1 — 12 unique failure-return lines (4 per file, `return { ok: false, reason: "exhausted_retries", ...};`) in 0/30 views; E3 — marker prints raw tool-output line numbers (21..198/199) while the view displays content line numbers (envelope offset 3); in content coordinates the declared range abuts head and tail exactly (no undeclared omission). Governance: R1 semantic diagnostic protection; the deployed keyword caliber is *not* retro-declared compliant; independent human review remains pending.

## S-F: P2 repair design (summary; design only, no runtime change)

Trigger analysis: the deployed DIAG pattern (`fail|error|assert|traceback|exception|fatal|denied|not found`) misses failure returns without those keywords. Repair design: (i) a three-layer deterministic detector (keyword layer; a structure layer matching failure-return shapes such as `ok:false/reason:/throw`; a context layer grouping catch blocks with their failure returns); (ii) dual-coordinate markers (raw and content line ranges plus envelope size); (iii) negative/regression/protection test matrix (old version must fail T-N1; byte-exact invariants and DIAG retention must hold; over-retention guard); (iv) expected impact: per-view character savings drop ≈1.5–3% on the audited corpus; token economics must be re-measured online for any repaired mechanism.

## S-G: Claim audit table (C1–C15, with evidence pointers)

| ID | Claim | Grade |
|---|---|---|
| C1 | No net input gain (3 mechanisms, 170 cases) | SS |
| C2 | Turn re-send amplification dominates character savings | SS |
| C3 | Safe surface ≈1.7% of context, below measurability | SS |
| C4 | No correctness regression (observed; not powered) | SS |
| C5 | Family structure (mutation+ / review−) | SWL |
| C6 | No runtime feature separates recovery need | SWL |
| C7 | Offline census predicts exposure | SS |
| C8 | $\hat{\sigma}_w = 0.155$ $[0.093, 0.197]$ | SS |
| C9 | Recovery is real agent behavior; chain complete | SS |
| C10 | Zero-recovery subgroup positive but not routable | SWL |
| C11 | Exposure generalizes (second corpus 29.5%, offline) | SWL |
| C12 | Turn-neutral silent truncation might work — only under an explicit diagnostic-visibility guarantee (absent in deployed M3) | H |
| C13 | Semantic-fidelity compression viable — hypothesis only; the deployed keyword-caliber protection did not achieve semantic diagnostic visibility | H |
| C14 | "All compression is useless" | U |
| C15 | "RDS +7.38% benefit" (D3-E5 degraded caliber) | U |

Evidence pointers: C1/C4 — three paired result sets + verification files; C2 — G4 decomposition + turn amplification; C3/C7 — grid + census; C8 — Stage-0 bootstrap; C9 — R6 screen + replay; C11 — D3 census; C12/C13 — G5 case study; C14/C15 — disavowed overreach formulations. Artifact roots: `r4/`, `r5/`, `paper-b/` (all hash-frozen).

## S-H: Number → evidence map (main-text numbers)

- M1 +7.1% (totals): `r4/100case/verification-and-analysis.json` primary_input_reduction −0.0708.
- M1 case mean −41.8% [CI]: recomputed from `r4/100case/paired-execution-results.jsonl` (script `scripts/fp_phase_a.py`; cross-checked to frozen mean-of-ratios convention).
- M1 542 folds / 561,454 chars / 64 cases: `rds_telemetry_T` in the same file.
- M2 −2.9% totals / 2 fold events: `r4/refinement/verification-and-analysis.json` (+0.0289; rds_T).
- M2 flow median −0.25%, tool ratio 1.02: same file (paired_flow_delta; tool_calls).
- M3 −23.0% [CI −40.8, −7.6] / 280 markers / 2.81M chars / 123 recoveries / 18→20: `r5/expansion/expansion-analysis.`
- +4.7 calls, +2.4 turns, +48% wall, +50.5K flow, +2,577 input, +46,265 cache: `expansion-analysis.json` totals + per-case fields.

- $\hat{\sigma}_w$ 0.155 [0.093, 0.197]; arms 0.258/0.244: `paper-b/draft-v1/step2-strengthening.json` S1/S2.
- Grid 0/64; max 1.66%; 245 blocked: `r4/refinement-r2/offline-verdict-report.md` (regenerated figure validated: 1.6586%).
- R6 311/43.4%/median 4/352,804: `r5/workload-screening-r1/step1-summary.json`.
- Feature bands 38–52%: `step23-feature-net.json`.
- D3 52,719 triggers / 366M chars / 29.5% of 1.239B: `step2-strengthening.json` S4.
- G4 S/R/M/X = 650,693 / 158,880 / 40,600 / 528,537; dual-caliber reconciliation: `r5/expansion/g4-cost-decompositio` `g4-final-metrology.json`, `phase0-consistency.json` P3.
- $r(\Delta\text{cache},\Delta\text{input}) = 0.33$; re-injected chars vs. $\Delta$input $r = 0.03$: recomputed from `expansion-analysis.json` cases (frozen inputs); corrects a misattributed $r = 0.34$ in the prior revision.
- G5 30/30 replay, 12 unique lines, offset 3: `paper-b/presubmission/g5-diagnostic-risk-rootcause-r1/` (rc-step1-reconstruction.json; rc-step2-contract-audit.json).